

Workshop 2: Keys and normalization

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In this workshop, you will work in a team on a table with many other students around keys and normalization in databases. It is a collective discussion to find a solution to the given problem, step by step by exchange and comments. You need paper and pens or "draw.io" to draw the Entity-Relationship Diagram (ERD) and a spreadsheet (Excel?) to create the tables. Please be positive, let all the people speak around the table, do not speak too loud, accept the comments for others, even negative comments can be treated as a way to improve your solution (always be positive). You can ask questions to me at any moment during the session. Välkomna, welcome, bienvenue...

1 Be positive! (5 min.)

Because, perhaps, you don't know each other around the table, take one minute per person to present yourself, explain who you are, where you come from, why you are taking this course/program and what you want to do in your future professional life. If everyone around the table is speaking Swedish, you can do it in Swedish, else, please go back to English.

2 Keys and Functional Dependencies (10 min.)

We consider the entity Person with the following attributes: persid, firstname, lastname, birthdate, email and phone. We know that the ID is a key for this entity, meaning that it uniquely identifies a person.

If we draw the functional dependencies between the attributes, because persid gives all the information, it has a functional dependency (FD) to all the attributes, so we obtain:



Discuss with your colleagues around the table about the following questions (and please answer them together):

- Are there any other functional dependencies than the arrows in the diagram? If yes, which ones?
- If you want the final table(s) to be in 3NF, what must you do?

- Draw the needed tables in 3NF to represent a Person, give an example of at least 3 records in each table.

3 Extracting table directly from dependencies (10 min.)

We are given an excel sheet in Table 1 that contains information about cars and persons, and we want to extract the tables from it. The excel sheet is the following:

<u>pid</u>	fname	lname	<u>cid</u>	plate	brand	model	email
17	Jérôme	Landré	15	ABC 123	Citroën	C3	jerome.landre@ju.se
19	Mira	Pop	17	DEF 456	Volvo	V90	mira.pop@ju.se
33	John	Doe	45	GHJ 789	BMW	X5	john.doe@ju.se
17	Jérôme	Landré	20	XYZ 012	Seat	600	jerome.landre@ju.se

Table 1: Information about persons and cars

- With your colleagues around the table, **discuss and draw** the functional dependencies for this table.
- When done, discuss and draw the tables in 3NF that you can extract from this excel sheet.

Tables:

4 Fika break! (15 min.)

It is time to take a 15 minute pause, go outside, walk, take a coffee, relax. And be positive: B+.

5 Closure (10 min.)

The closure $\{A\}^+$ is the set of all attributes you can reach directly or indirectly from $\{A\}$. The attribute itself is always in its closure (from attribute A, you can reach at least attribute A).

Given the functional dependencies in the following form:

$\{A\} \rightarrow \{B, C\}$

$\{C\} \rightarrow \{G\}$

$\{E\} \rightarrow \{F, H, I, J\}$

$\{H\} \rightarrow \{J, K, L\}$

- Draw the diagram with the functional dependencies represented as arrows.

A B C D E F G H I J K L

- Compute the closures:

$\{A\}^+ =$

$\{B\}^+ =$

$\{C\}^+ =$

{D}+ =

{E}+ =

{F}+ =

{G}+ =

{H}+ =

{I}+ =

{J}+ =

{K}+ =

{L}+ =

- From these closures, can you find the tables and the keys for them? Explain your answer.

6 Music (20 min.)

By studying the given the information in Table 2, can you find the functional dependencies and the tables in 3NF that you can extract from it?

AlbumID	AlbumName	ArtistName	ArtistBio	ReleaseYear	SongTitle	SongDuration
1	Thriller	Michael Jackson	King of Pop, born 1958	1982	Billie Jean	4:54
1	Thriller	Michael Jackson	King of Pop, born 1958	1982	Beat It	4:18
2	The Dark Side of the Moon	Pink Floyd	British rock band, formed 1965	1973	Time	7:06
2	The Dark Side of the Moon	Pink Floyd	British rock band, formed 1965	1973	Money	6:22

Table 2: Music.

- Discuss your solution with your colleagues.

7 If you have finished...

In case you have finished the workshop very fast, you can work as a group on the lab assignment. I know you will also work with your partner on it, but it can help to start drawing some parts of the ERD by discussing with your colleagues around the table. :) B+